

Dr. Babasaheb Ambedkar Technological University, Lonere

Regular And Supplementary Summer 2024

Course: B. Tech.

Branch: Civil Engineering

Semester: VIII

Subject Code: BTCESS802A

Subject Name: Energy Efficiency Acoustics and Day lighting in Building

Max Marks: 60

Date: 15/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/ CO)	Marks
Q. 1 Solve Any <u>Two</u> of the following.		12
A) Discuss in brief- Impact of environment on building	CO1	06
B) What is Green building? Explain in detail.	CO1	06
C) How the construction of building affects adversely on environment?	CO1	06
Q.2 Solve Any <u>Two</u> of the following.		12
A) State factors affecting on human comfort.	CO2	06
B) Explain in detail human response to thermal environment.	CO2	06
C) Explain Heat Flow in buildings.	CO2	06
Q. 3 Solve Any <u>Two</u> of the following.		12
A) What is Admittance Method?	CO2	06
B) State effect of solar radiation.	CO2	06
C) Explain in detail Thermal properties of material and sections	CO2	06
Q.4 Solve Any <u>Two</u> of the following.		12
A) Explain Mechanisms of Natural ventilation	CO2	06
B) How to increase natural ventilation in a building?	CO2	06
C) Write a short note on Artificial Lighting.	CO3	06
Q. 5 Solve Any <u>Two</u> of the following.		12
A) Write a short note on Basic acoustics and noise	CO3	06
B) What are the economic benefits of green or sustainable building and development?	CO3	06
C) Explain in detail Noise Outdoors.	CO3	06

*** End ***